EasyCGTree

An Easy Tool for Constructing Core-Gene Tree

Version 5.0

with EasyCGTree_GUI 1.0

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Updated on October 9, 2026

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1. What is EasyCGTree?

EasyCGTree is a pipeline, developed to construct **genome-based Maximum-Likelihood (ML) phylogenetic trees**, by taking microbial genomic data (amino acid and/or DNA sequences) in FASTA format as input data. Profile hidden Markov models (HMMs) of core gene sets prepared in advance and enclosed in the package are used for homolog search by HMMER (http://hmmer.org/)[1], and customized gene sets (prepared as gene clusters, which means homologous gene sequences in one file) can also be used to build profile HMMs for homolog search. EasyCGTree includes three approaches to infer phylogeny: **supermatrix**, **supertree**[2] and **consensus tree**. It has integrated all the steps required between the input genomic data and the resulting tree file, which makes it easy to infer a core-gene tree. Furthermore, the intermediate data of an EasyCGTree run can be directly used as input data of many other applications.

Version 5.0 brings the following main changes:

1) **EasyCGTree_GUI**, a graphical interface for Windows, Linux and macOS (no command line needed). It shows the commands it runs, so they can also be copied and used in a terminal.

2) **macOS is supported.** Ready-to-use packages are provided for Windows, Linux and macOS (Apple silicon and Intel) on GitHub and Gitee; the HMM sets are downloaded separately as needed.

3) A separate CDS prediction task (`-task predict`), so several HMM sets can be tried without predicting the CDS again; NCBI annotation files (protein and nucleotide sequences of the CDS) can be used directly as input data.

4) Trees from **nucleotide** alignments of the core genes (`-seq nucl`), and **SNP analyses**, which extend EasyCGTree to the level below species: core-genome SNPs and SNP trees (EasyCGTree_SNP.pl), and strain- and clade-specific SNPs on a tree (EasyCGTree_SpecificSNP.pl).

5) All results can be written to any folder (`-outdir`); the scripts can be called from any directory; the settings of each task are recorded, so later tasks check that they fit the earlier results, which avoids unnecessary repeated computation.

6) Up-to-date programs: HMMER 3.4, MUSCLE v5, IQ-TREE 3 and others.

1.1 Key terms

Some terms used in this manual, explained briefly for new users:

Term	Meaning
Core genes	Genes present in (nearly) all genomes of a group, usually in a single copy. Their sequences are compared to infer the tree.
Profile HMM / HMM set	A statistical model of a gene family, built from many sequences of the family. HMMER uses it to find the gene in a genome, even if the sequence differs considerably. An HMM set (e.g. bac120) contains one model per core gene.
E-value	Number of hits with at least this score expected by chance; the smaller, the more reliable a hit. Hits with an E-value above '-evalue' are ignored.
Genome / gene cutoff	Screening of the data: genomes in which too few genes were found, and genes found in too few genomes, are left out (Section 5.3).
Alignment / trimming	The sequences of a gene are aligned so that homologous positions are in the same column. Trimming removes unreliable columns (e.g. with many gaps).
Supermatrix (sm)	The alignments of all genes are joined into one long alignment, and one tree is inferred from it. Fast and robust; the default.
Supertree (st) / consensus (cs)	A tree is inferred for each gene, and the gene trees are combined (st: by ASTRAL; cs:

Term	Meaning
	majority-rule consensus) to estimate the species tree. It also shows how much the genes agree.
ML tree	Maximum-likelihood tree: the tree under which the observed alignment is most probable for a model of sequence evolution (FastTree, IQ-TREE).
Support value	A number at each branch (e.g. bootstrap or SH-like support, 0-1 or 0-100) telling how strongly the data support this group. Low values mean an uncertain branch.
Newick	The common text format of trees, e.g. '((A,B),C);'. It can be opened with FigTree, MEGA, iTOL and others.
Root / outgroup	Inference programs give unrooted trees. The root (the oldest point of the tree) is placed by an outgroup: strains known to be outside the group of interest (Section 7.2).
SNP / core SNP	Single nucleotide polymorphism: an alignment column in which the strains have different bases. Core SNPs are SNP sites present (without gaps) in all strains.
Threads	Number of processor cores used at the same time; more threads make most steps faster.

2. How to install EasyCGTree?

2.1 Download

Ready-to-use packages of EasyCGTree are provided on the **Releases** page of the repository **EasyCGTree5** on GitHub (https://github.com/zdf1987/EasyCGTree5/releases) and on Gitee (https://gitee.com/zdf1987/EasyCGTree5/releases; recommended in mainland China). Download the package for your operating system:

Operating system	Package	Graphical interface
Windows 10/11 (64-bit)	`EasyCGTree5-Windows-x64.zip`	`EasyCGTree_GUI.exe`
Linux (64-bit, glibc 2.35 or newer)	`EasyCGTree5-Linux-x64.tar.gz`	`EasyCGTree_GUI`
macOS 11 or newer, Apple silicon (M1, M2 ...)	`EasyCGTree5-macOS-arm64.zip`	`EasyCGTree_GUI.app`
macOS 11 or newer, Intel	`EasyCGTree5-macOS-x64.zip`	`EasyCGTree_GUI.app`

Decompress the package into a folder whose path contains **no spaces and no non-English characters** (e.g. `D:\EasyCGTree5` for Windows, `~/EasyCGTree5` for Linux and macOS). No installation is required for EasyCGTree itself; it runs on any system with Perl. The folder obtained is called the 'home directory' of EasyCGTree in this manual.

2.2 Contents of the package

```
EasyCGTree5/
  EasyCGTree_GUI(.exe/.app)   graphical interface (Section 4)
  EasyCGTree.pl               main pipeline (Section 5)
  EasyCGTree_SNP.pl           SNPs and SNP tree (Section 7.1)
  EasyCGTree_SpecificSNP.pl   specific SNPs (Section 7.2)
  BuildHMM.pl                 builds profile HMMs (Section 7.3)
  Gene_Prevelence.pl          gene distribution (Section 7.4)
  bin/                        programs, tree_app-options.txt
  HMM/                        README.txt only; HMM sets are downloaded
                              separately (Section 3.2)
```

2.3 Perl and the programs used by EasyCGTree

The scripts of EasyCGTree are written in Perl and use only its core modules. **Perl** (5.14 or newer) is already installed on Linux and macOS. **Windows users need to install Perl**, e.g. Strawberry Perl (https://strawberryperl.com/); the installer adds Perl to the system path, and EasyCGTree_GUI finds it automatically.

The other programs invoked by EasyCGTree are included in the folder 'bin' of each package:

1) HMMER[1] (v3.4): http://hmmer.org/
2) Prodigal[3] (v2.6.3): https://github.com/hyattpd/Prodigal
3) MUSCLE[4] (v5.3): https://www.drive5.com/muscle/
4) trimAl[5] (v1.5.1): http://trimal.cgenomics.org/
5) FastTree[6] (v2.2, multi-threaded FastTreeMP): https://morgannprice.github.io/fasttree/
6) IQ-TREE[7] (v3.1.4): https://iqtree.github.io/
7) ASTRAL (wASTRAL[8] of ASTER v1.25, as 'astral-weighted'): https://github.com/chaoszhang/ASTER

Linux and macOS: if a program in 'bin' cannot be started ('Permission denied'), make the programs executable with `chmod +x bin/*` in the home directory of EasyCGTree. macOS may also block programs downloaded from the internet; run `xattr -dr com.apple.quarantine .` in the home directory of EasyCGTree once (see Section 9).

Experienced users can replace the included programs with other versions: copy them into 'bin' with the same names, or set their paths in EasyCGTree_GUI (Settings → Paths and programs, Section 4.6).

2.4 First start

Start **EasyCGTree_GUI** by double-clicking it (Linux: `./EasyCGTree_GUI` in a terminal also works). EasyCGTree_GUI finds the scripts, 'bin' and 'HMM' in its own folder. Then choose **Help → Check installation**: Perl, the five scripts, every program in 'bin' (with its version) and the HMM sets are tested, and problems are listed with hints.

Before the first run, download at least the default HMM set 'bac120.hmm' into the folder 'HMM' (Section 3.2).

Windows: SmartScreen may warn about an unknown program at the first start ('More info' → 'Run anyway'). macOS: the app is not signed by Apple; right-click EasyCGTree_GUI.app and choose 'Open' at the first start.

2.5 Run or build EasyCGTree_GUI from the source code (optional)

The packages contain EasyCGTree_GUI as a ready-to-use program. Users who prefer the source code (e.g. on a system for which no package is provided) can run it with Python, or build the program themselves. The source code (`EasyCGTree_GUI.py`) and the build scripts (folder `build`) are in the GitHub repository.

Run with Python

Python 3.9 or newer is needed. Install the Qt library PySide6 once, then start EasyCGTree_GUI.py (put it into the home directory of EasyCGTree, next to EasyCGTree.pl):

```
$ python3 -m pip install PySide6
$ python3 EasyCGTree_GUI.py          (Windows: py EasyCGTree_GUI.py)
```

Build the program

The program is made with PyInstaller. A program can only be built on the operating system it is made for (the Windows program on Windows, the macOS app on a Mac). In the folder `build`, run the script for your system; Python and an internet connection are needed only for building:

System	Command	Result
Windows	double-click `build_windows.bat`	`build\dist\EasyCGTree_GUI.exe`
macOS	`bash build_macos.sh`	`build/dist/EasyCGTree_GUI.app`
Linux	`bash build_linux.sh`	`build/dist/EasyCGTree_GUI`

Each script installs PySide6 and PyInstaller into a separate environment (`.venv`) and runs a command like:

```
pyinstaller --onefile --windowed --name EasyCGTree_GUI ../EasyCGTree_GUI.py
```

Copy the result into the home directory of EasyCGTree. On GitHub, the workflow `.github/workflows/build-gui.yml` of the repository builds the programs for all systems at once, and `build-release.yml` builds the complete packages (see `build/README_BUILD.txt`).

3. Input data

Genomic data in FASTA format (**protein sequences of CDS, nucleotide sequences of CDS or genomic DNA sequences, alone or mixed**) are enough to run EasyCGTree. To use a customized gene set for homolog search, prepare the gene clusters and build the profile HMM with BuildHMM.pl first.

3.1 Genomic data (FASTA format)

Gather the genomic data into one folder. EasyCGTree judges the type of each file and, where necessary, **predicts the CDS of DNA sequences with Prodigal**. The FASTA files need to be uncompressed. File names and sequences are prepared automatically (spaces, special characters, lower-case sequences ...): **the names of the files without extension become the labels of the tree tips**. With 'Check input' in EasyCGTree_GUI (or `-dry_run`), users can see how the files will be grouped and named before running anything.

CDS sequences provided by users are treated as follows:

a. NCBI protein file with locus_tag (`translated_cds.faa`) + CDS nucleotide file (`cds_from_genomic.fna`) of the same assembly (the file names must contain the assembly accession, e.g. `GCF_000009925.1_ASM992v1_...`): the sequences are paired by locus_tag and both are used directly. If fewer than 90% of the proteins can be paired, the two files probably do not belong to the same genome, and EasyCGTree stops with an error. These genomes are named by their accession (e.g. `GCF-000009925-1`).

b. Protein sequences of CDS alone: used directly, but analyses of nucleotide sequences are not possible. NCBI nucleotide sequences of CDS alone (sequence names containing `[locus_tag=...]`, e.g. `cds_from_genomic.fna`): the CDS are translated into proteins (genetic code 11).

c. If the genome (`genomic.fna`) and NCBI CDS sequences of the same assembly are both provided, the annotation files (CDS) are used and the genome is ignored.

Paired protein and nucleotide files from other sources cannot be checked reliably, so each file is treated as a genome of its own; nucleotide CDS sequences without locus_tag are treated as genomic DNA, and their CDS are predicted again by Prodigal. For data not from NCBI, please provide only one file per genome (e.g. the genomic DNA sequences).

Trees from nucleotide sequences (`-seq nucl`) and the SNP analyses (Section 7) require all genomes to be DNA sequences (genomes or nucleotide CDS files).

3.2 Profile HMM

The set of genes used to build the profile HMM determines which genes will be extracted from the proteomes to infer a core-gene tree. How to define the core genes of the data set of interest? The best way is to define the pan-genome first. Nevertheless, if a researcher just wants a core-gene tree, a pan-genome analysis is not the optimal way; retrieving a core gene set from previous studies or public databases may be a better choice.

EasyCGTree provides profile HMMs of several core gene sets prepared in advance, which are referred to as the profile HMM database (PHD). Because the HMM files are large (about 220 MB together) and most users need only a few of them, they are not included in the program package: the folder 'HMM' of the package contains only 'README.txt'. Download each set you need from the folder 'HMM' of the repository and save it, without changing its name, in the folder 'HMM' of EasyCGTree:

GitHub: https://github.com/zdf1987/EasyCGTree5/tree/main/HMM

Gitee: https://gitee.com/zdf1987/EasyCGTree5/tree/main/HMM (recommended in mainland China)

To download one file, open it in the folder 'HMM' of the repository and click 'Download raw file' (GitHub) or 'Download' (Gitee); in EasyCGTree_GUI, **Help → HMM sets online** opens these pages. The default set is bac120: download 'bac120.hmm' to run EasyCGTree with its default settings. If a set is missing, EasyCGTree and EasyCGTree_GUI stop with a message that points to 'HMM/README.txt' and to the repositories. The sets available are (details and references in 'HMM/README.txt'):

Group	Gene set (-hmm)	Description
Prokaryotes	rp1	16 ubiquitous ribosomal protein genes (18 domains)
	rp2	23 ubiquitous ribosomal protein genes (27 domains)
Bacteria	bac120	120 ubiquitous genes in the domain *Bacteria* (default)
	essential	107 essential single-copy core genes
	ubcg1 / ubcg2	92 / 81 up-to-date bacterial core genes (UBCG)
	bac52	52 ubiquitous ribosomal protein genes
Archaea	ar122	122 ubiquitous genes in the domain *Archaea*
	uacg	128 ubiquitous genes in the domain *Archaea*
	ar53	53 ubiquitous ribosomal protein genes
Others	ery288 / rhodo268 / spi269	core genes of *Erythrobacteraceae* / *Rhodobacteraceae* / *Spirosomaceae*
	CKC20	20 core genes of the *Corynebacterium kroppenstedtii* complex
	AXT	5 key genes of astaxanthin synthesis

The PHD is extensible and can be customized: gene sets (prepared as gene clusters) supplied by users can be used to build profile HMMs (EasyCGTree_GUI tab 'HMM building', or BuildHMM.pl, Section 7.3). A .hmm file elsewhere can also be used directly.

4. Run EasyCGTree with the graphical interface

4.1 The main window

EasyCGTree_GUI has three tabs: **Main pipeline** (EasyCGTree.pl and Gene_Prevelence.pl), **SNP analysis** (EasyCGTree_SNP.pl and EasyCGTree_SpecificSNP.pl) and **HMM building** (BuildHMM.pl). Each tab has the same layout of four panels (Fig. 1):

a. **Parameters** (top left): global settings first, then the steps in the order in which they are run. Options not used by the chosen tasks are greyed out; '?' next to an option explains it.

b. **Status** and **Files** (top right): two pages of one panel, chosen at the top of the panel. The buttons 'Output' and 'Input' next to them choose the folder shown on both pages. For the output directory, Status shows what has been done (prediction, HMM search, screening, alignments, trees); for the input directory, the genomes, their sequence types and the number and length of their sequences. In Files, double-click opens folders, tables (in the GUI) and other files.

c. **Command line** (bottom left): one row with 'Concise'/'Full', 'Copy', 'Run' and 'Stop', and below it the commands that 'Run' will execute. 'Concise' leaves out default values, 'Full' shows all options; 'Copy' copies them for a terminal.

d. **Run log** (bottom right): the state of the run (running, finished, failed or stopped) next to the title, and the messages of the running scripts. 'Stop' ends a run, including all programs started by it.

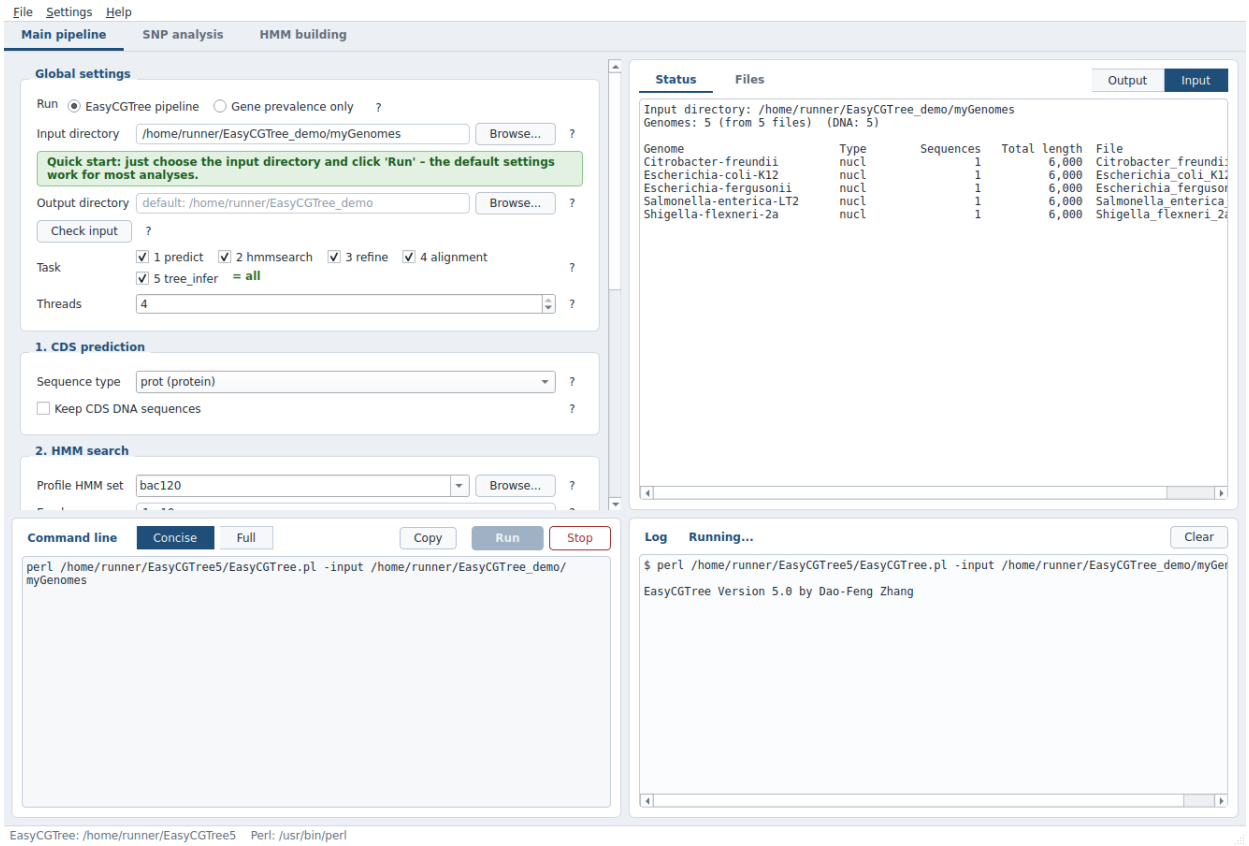


Fig. 1 The main window of EasyCGTree_GUI

Drag the bars between the panels to change the heights of the top and bottom halves and the widths of the two panels in each half. In a small window, show the page you need in the top-right panel or widen the parameter panel.

The language (English/Chinese) is chosen in **Settings → Language**. Paths and programs, the language, and the sizes of the window and the panels are kept for the next start; the analysis parameters start with their default values each time (they can be saved and loaded with **File → Save/Load parameters**).

4.2 Main pipeline: quick start

1) Click 'Browse...' next to **Input directory** and choose the folder with the genomic data (Section 3.1).

2) Optionally, choose an **Output directory**; by default the results are written to the folder that contains the input directory.

3) Click **Run**.

With the default settings, a supermatrix tree of the bac120 genes is inferred with FastTree from protein sequences (e.g. `myGenomes.bac120.119.supermatrix.fasttree.tree`, Section 6). 'Check input' shows how the input files will be grouped and named without running anything.

4.3 Main pipeline: tasks and options

Task: the five tasks 1 predict, 2 hmmsearch, 3 refine, 4 alignment and 5 tree_infer (Fig. 2) can be ticked. They must follow each other: ticking two tasks also ticks the tasks between them. All five ticked means a complete run ('all'); otherwise, one command per task is run, one after the other. If a CDS prediction of the same input files with the same settings exists already, a yellow notice is shown and 'predict' is unticked, so the existing prediction is reused; close the notice to tick 'predict' again.

The other options correspond to those of EasyCGTree.pl (Section 5.2) and are grouped by step: 1. CDS prediction (sequence type, keeping the CDS DNA sequences), 2. HMM search (HMM set, E-value), 3. Screening (genome and gene cutoffs), 4. Alignment and trimming (trimAl method), 5. Tree inference (approach and program). The number of threads is limited to the number of CPU cores of the computer.

Gene prevalence: 'Summarize afterwards' runs Gene_Prevelence.pl after the pipeline; with the run mode 'Gene prevalence only', only the distribution of the genes is summarized (Section 7.4).

4.4 SNP analysis

The tab runs up to three steps for the genomes of an input directory (all genomes must be DNA sequences):

1) **SNP extraction** (EasyCGTree_SNP.pl): if the nucleotide alignments of the core genes are missing, the required EasyCGTree tasks are run first with '-seq nucl'. The EasyCGTree settings of the tab are filled in from the record of the previous run (or the defaults).

2) **SNP tree** with IQ-TREE or FastTree.

3) **Strain- and clade-specific SNPs** (EasyCGTree_SpecificSNP.pl), counted on the SNP tree or on another tree ('Other tree', e.g. the supermatrix tree; a Newick tree whose strain names are those of the SNP alignment). With another tree, step 2 is not needed and step 1 is run only if needed. For the interpretation of clade-specific SNPs, the tree must be rooted correctly; give the outgroup strains in 'Outgroup' (see Section 7.2).

Ignored strains, the reference strain and the outgroup are chosen from a list of the genomes ('Choose...'). If the settings differ from the previous run, a notice tells which EasyCGTree step is run again and why. Result files with the same name are not overwritten but renamed to `<file>.<date>.bak`.

4.5 HMM building

Choose the **Gene family directory** (one FASTA file of protein sequences per gene family, e.g. the folder 'TEM4_GeneCluster' of an EasyCGTree run) and click 'Run'. The new HMM set is written to the folder 'HMM' with

the name given in 'Name of the HMM set' (filled in from the folder name) and is then available in the other tabs. With 'Already aligned', aligned families are not aligned again.

4.6 Settings

a. **Paths and programs**: the home directory of EasyCGTree, the 'bin' directory, Perl, and the path of each program. A program set here replaces the one in 'bin' (e.g. an IQ-TREE installed elsewhere).
b. **Check installation** (menu Help): tests Perl, the scripts, the programs and the HMM sets.
c. **HMM sets online** (menu Help): opens the folder 'HMM' of the repository on GitHub or Gitee, or the local folder 'HMM'.
d. **Tree program command lines**: edits 'bin/tree_app-options.txt' (options of IQ-TREE, FastTree and ASTRAL, Section 5.4).

5. Run EasyCGTree on the command line

5.1 Usage

All functions of EasyCGTree_GUI are also available on the command line (**cmd.exe** or PowerShell on Windows, **Terminal** on Linux and macOS). The scripts can be called from any directory, and the input directory can be anywhere. Assume that EasyCGTree was decompressed to `D:\EasyCGTree5` (Windows) or `~/EasyCGTree5` (Linux/macOS), and the genomes are in the folder `myGenomes`:

```
> perl D:\EasyCGTree5\EasyCGTree.pl -input D:\data\myGenomes
```

```
$ perl ~/EasyCGTree5/EasyCGTree.pl -input ~/data/myGenomes -tree st -tree_app iqtree -
thread 8
```

Then a tree (named after the input folder, e.g. `myGenomes.bac120.119.supermatrix.fasttree.tree`) in Newick format is written next to the input folder (or into `-outdir`), together with a log file and the working directory `myGenomes_TEM` (Section 6). The command lines shown by EasyCGTree_GUI (Section 4.1) can be copied and run in the same way.

Note: Windows uses '\' to separate folders, while Linux and macOS use '/'. A path containing spaces must be put in quotes.

5.2 Options of EasyCGTree.pl (in any order)

-input <String> **(The sole essential option)**

Input directory including all genomic data sets (Section 3.1).

-outdir <String>

Output directory for all results (working directory '<input>_TEM', trees, log files). [default: the directory that contains the input directory]

A later task must be given the same '-outdir' as the earlier ones; otherwise their results are not found: a later task run alone stops with an error, and a complete run starts again in the new place.

-task <String, 'all', 'predict', 'hmmsearch', 'refine', 'alignment', 'tree_infer'>

Set run mode (Fig. 2). [default: all]

Each task uses the results of the earlier tasks in '<input>_TEM'; the settings of each task are recorded there, and a later task checks that they fit (e.g. the same HMM set).

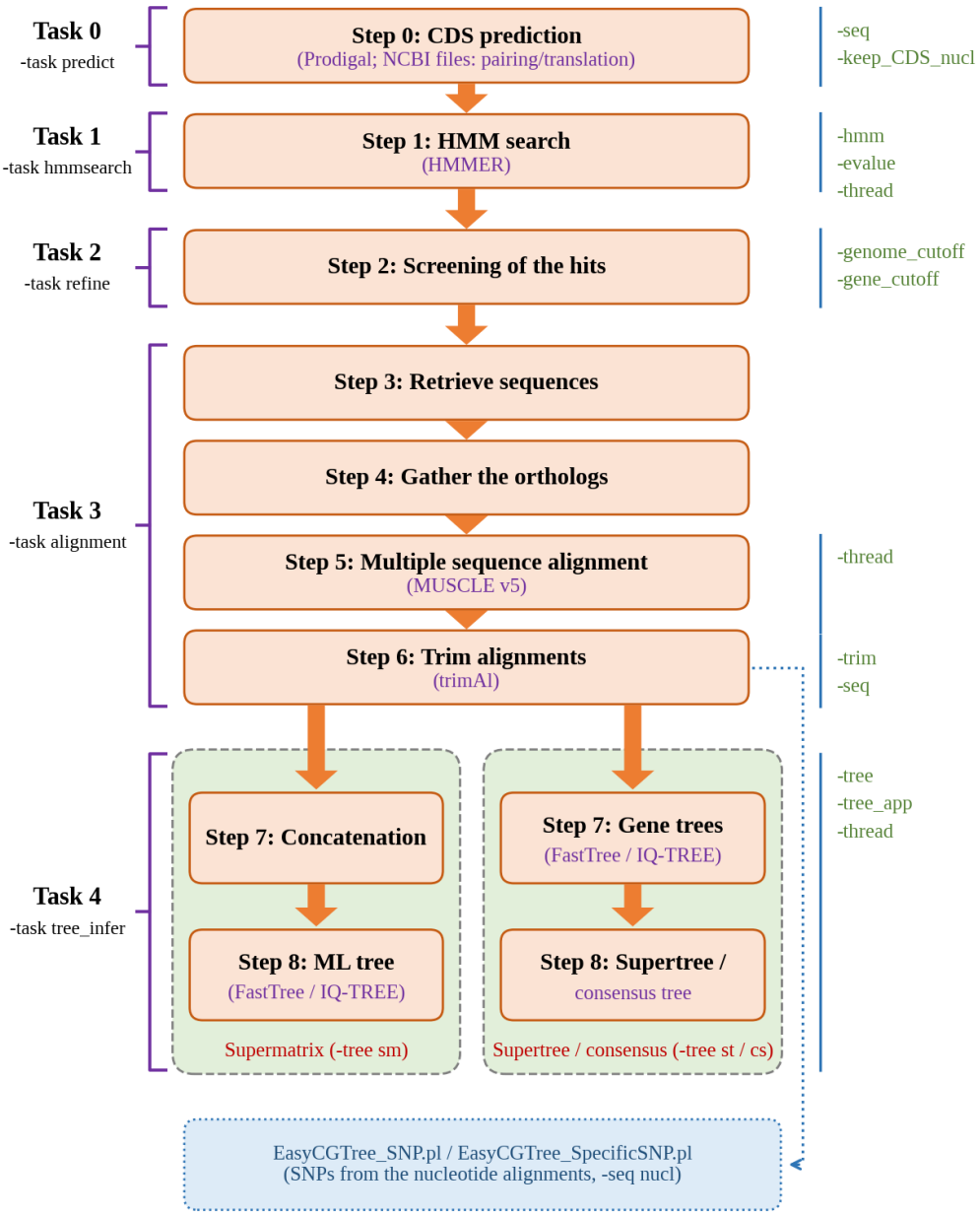


Fig. 2 Program flowchart of EasyCGTree for a complete run

-seq <String, 'prot', 'nucl'>

Sequence type (amino acid or nucleotide) used for tree inference. [default: prot]

'nucl' requires all genomes to be DNA sequences.

-keep_CDS_nucl

Also keep the CDS DNA sequences of the genomes provided as DNA (e.g. to build nucleotide alignments later). Not needed with '-seq nucl'.

-hmm <String, 'bac120', 'rp1', et al., or a .hmm file>

Profile HMM used for gene calling (Section 3.2). [default: bac120]

-evalue <Real, 10..0>

Expect value for saving hits during the HMM search. [default: 1e-10]

-genome_cutoff <Decimal, 0..1>

Cutoff for omitting low-quality genomes. [default: 0.8]

-gene_cutoff <Decimal, 0..1>

Cutoff for omitting low-prevalence genes. [default: 0.8]

With '-tree cs', 1 is always used; with '-tree st', 1 is recommended.

-trim <String, 'gappyout', 'strict', 'strictplus', 'nogaps'>

Standard for trimming alignments used by trimAl. [default: strict]

'nogaps' removes every column that contains a gap; genes without any column left are skipped.

-tree <String, 'sm', 'st', 'cs'>

Approach used for tree inference. [default: sm]

sm, supermatrix (concatenated alignment); st, supertree (gene trees combined by ASTRAL); cs, majority-rule consensus of the gene trees (by IQ-TREE).

-tree_app <String, 'fasttree', 'iqtree'>

Application used for tree inference. [default: fasttree]

-thread <Int>

Number of threads used by HMMER, MUSCLE, FastTree, IQ-TREE and ASTRAL. [default: 4]

If it is larger than the number of CPU cores, the number of cores is used.

-dry_run

Only check the input directory: show how the input files would be grouped and named, without renaming or running anything.

-help

Display the help message.

Which settings should I choose?

a. **Defaults first.** The default settings (bac120, protein sequences, supermatrix, FastTree) give a good tree for most data sets of bacteria; for archaea, choose an archaeal HMM set (e.g. ar122 or ar53).

b. **Protein or nucleotide?** Protein sequences (`-seq prot`) are best for genera, families and higher ranks. For closely related strains (e.g. within a species), protein sequences show too few differences: use `-seq nucl`, or the SNP analysis (Section 7.1).

c. **FastTree or IQ-TREE?** FastTree is fast and good for a first look or large data sets. IQ-TREE selects the best substitution model and computes ultrafast bootstrap support; it is slower and needs more memory.

d. **Supermatrix, supertree or consensus?** The supermatrix (sm) is usually the most reliable and the fastest. Supertree (st) and consensus (cs) infer one tree per gene, which takes longer; they show whether the genes tell the same story (e.g. low consensus values indicate conflicting genes).

5.3 More comments on '-genome_cutoff' and '-gene_cutoff'

The two parameters may be confusing to some users. In most situations, leave them alone and the default settings will be OK for a run involving a limited number of genomes (<30), especially when the data are all from complete genomes. However, if many more genomes are used, it is inevitable that some genes are not detected in some genomes, and the number of common genes decreases. In fact, many genomes collected in the representative database of GTDB (Genome Taxonomy Database, https://gtdb.ecogenomic.org) contain an incomplete bac120/ar122 gene set. It is a trade-off between the number of common genes and the number of genomes.

The two parameters are expected to balance this by excluding low-prevalence genes and low-quality genomes. Users should be aware that these two parameters determine the floor level of the quality of input genomes (the selected gene set is determined by the selected genomes). Lower cutoffs make no sense if the genomes are all of high quality. For example, users will always get 120/122 genes to infer a tree if all the genomes contain a complete

bac120/ar122 gene set, no matter what is specified (0.5-1). Gene_Prevelence.pl (Section 7.4) previews how many genomes and genes pass given cutoffs.

5.4 Options of the tree programs

The command lines of IQ-TREE, FastTree and ASTRAL can be changed in **'bin/tree_app-options.txt'** (or in EasyCGTree_GUI: Settings → Tree program command lines), e.g. `IQ-TREE Command-line=iqtree3 -B 1000` for 1000 ultrafast bootstrap replicates. The input alignment and the number of threads are added by EasyCGTree.

6. Output files

Assume the input folder is 'myGenomes'.

1) Phylogenetic trees in Newick format. Users can display them by using FigTree, MEGA, iTOL or other tree viewers. The final tree file name looks like this:

```
myGenomes.bac120.119.supermatrix.fasttree.tree
   (1)     (2)   (3)     (4)       (5)
```

(1) name of the input folder (-input); (2) gene set used (-hmm); (3) number of genes used for tree inference; (4) approach (-tree: supermatrix, supertree or consensus); (5) program used (-tree_app: fasttree or iqtree). If a tree with the same name exists already, it is renamed to '<name>.<date>.bak' instead of being overwritten.

2) A log file (e.g. `myGenomes.bac120.sm.fasttree_2026-10-09-10-30.log`) named after the input folder, HMM, approach, program and the starting time (`myGenomes.predict_<time>.log` for '-task predict'). It records the details of the run, e.g. which genes and genomes were included or excluded, and the substitution model selected by IQ-TREE.

3) A folder (`myGenomes_TEM`) named after the input folder and ending with '_TEM'. The names of its contents are the same for all runs of EasyCGTree:

a. **TEM0_CDS:** proteins (.faa) and, if needed, CDS DNA sequences (.fnn) of each genome, copied from the input or predicted by Prodigal.

b. **TEM1_HMMsearch_out:** HMM search results of each genome.

c. **TEM2_HMMsearch_outS:** screened HMM search results: the best hit of each gene (E-value and score cutoffs).

d. **TEM3_GeneSeqs:** gene sequences retrieved from each genome.

e. **TEM4_GeneCluster:** one file per gene with the homologs of all genomes. These files can be used directly to build a profile HMM (Section 7.3).

f. **TEM5_Alignment** and **TEM6_AlnTrimmed:** alignments made by MUSCLE, and the same alignments trimmed by trimAl.

g. **TEM7_GeneTrees:** gene trees (approaches st and cs), and **intree:** all gene trees in one file, used by ASTRAL or IQ-TREE.

h. **<name>.concatenation.fas:** the concatenated alignment of the selected genes (approach sm). It can also be used by other programs, e.g. RAxML or FastME.

i. **TEM8_SNP:** working files of EasyCGTree_SNP.pl.

j. **Genome_SeqType.txt** (input type and files of each genome), **HMMinfo.txt** (genes and lengths of the HMMs), **GenomeGeneScreened.txt** (genomes and genes after screening), **trimAl_Details.txt** (alignment lengths before and after trimming), **EasyCGTree_record.txt** (settings and times of the tasks performed; please do not edit).

7. Other scripts

The four scripts are used like EasyCGTree.pl (they are run by EasyCGTree_GUI, too). Each of them writes a log file, and runs the required EasyCGTree tasks by itself if their results are missing. The common options `-outdir`, `-thread` and `-help` work as in EasyCGTree.pl.

7.1 EasyCGTree_SNP.pl

Extracts SNPs from the concatenated nucleotide alignments of the core genes and infers a SNP tree. If the nucleotide alignments are missing, or were made with other settings than the ones given, the required EasyCGTree tasks are run with '-seq nucl'.

```
> perl EasyCGTree_SNP.pl -input myGenomes [options]
```

-input <String> (Essential)
The input directory used (or to be used) with EasyCGTree. All genomes must be DNA sequences.

-ignore <String>
Strains ignored when SNP sites are detected (comma-separated names, or a file with one name per line). They stay in the SNP alignment and the tree, but sites that vary only because of them are not SNPs.

-max_missing <Decimal, 0..1>
Largest fraction of strains allowed to have a gap or an ambiguous base at a SNP site. [default: 0, i.e. core SNPs]

-aln <String, 'trimmed', 'original'>
Alignments used: trimmed by trimAl, or before trimming. [default: trimmed]

-ref <String>
Reference genome: the positions of the SNPs in its CDS and the codon positions are reported too.

-tree_app <String, 'iqtree', 'fasttree'> / -no_tree
Program for the SNP tree [default: iqtree]; '-no_tree': only extract the SNPs.

-asc <String, 'fconst', 'lewis'>
Correction of IQ-TREE for using only variable sites. [default: fconst]

-hmm, -evalue, -gene_cutoff, -genome_cutoff, -trim
Passed to EasyCGTree if its tasks have to be run [default: the settings of the previous run].

Output: `<input>.<hmm>.<genes>.snp.fas` (SNP alignment), `.snp_positions.txt` (position of each SNP), `.snp_genes.txt` (SNPs per gene), `.snp_distance.txt` (pairwise SNP distances) and `.snp.<tree_app>.tree` (SNP tree).

7.2 EasyCGTree_SpecificSNP.pl

Counts strain- and clade-specific SNPs on a tree: each strain name gets '_N' (N: SNPs specific to the strain), and each clade gets the label 'N/S' (N: SNPs specific to the clade; S: support value). The SNP results of EasyCGTree_SNP.pl are used (it is run first if they are missing).

```
> perl EasyCGTree_SpecificSNP.pl -input myGenomes -outgroup strainA,strainB [options]
```

-tree_app <String, 'iqtree', 'fasttree'>
SNP tree to annotate. [default: iqtree]

-tree <String>

Annotate another tree instead (e.g. the supermatrix tree of EasyCGTree): a Newick tree whose strain names are exactly those of the SNP alignment. It should be rooted as wanted, or the outgroup is given with '-outgroup'.

-outgroup <String>

Root the tree with these strains (comma-separated; they must form one clade).

-mode <String, 'exclusive', 'strict'>

exclusive: all strains of the group share a base that no other strain has [default]; strict: in addition, all other strains share one base (sites without missing data only; independent of the rooting).

Output: `<out>.tree` (tree with the numbers of specific SNPs), `<out>.clade_ID.tree` (clade IDs as labels), `<out>.positions.txt` (each specific SNP) and `<out>.groups.txt` (each strain and clade: support, SNPs, members).

Why the root and the outgroup matter

A clade-specific SNP is a base shared by all strains of a clade and by no other strain. Which strains form a clade, however, depends on where the root of the tree is. Trees inferred by IQ-TREE and FastTree are **unrooted**: they only tell which strains are separated by each branch, and the root written in the file is arbitrary. Every branch splits the strains into two sides; the side that is called 'the clade' is the side away from the root. With a wrong root, a clade can be turned 'inside out' (the SNPs of a group are reported for the rest of the tree), and groups that are not real clades appear. Therefore the tree should be rooted correctly before the clade-specific SNPs are interpreted.

Choosing an outgroup: use one or a few strains that are certainly outside the group of interest but not too distant, e.g. strains of a closely related species of the same genus. Include them in the input folder from the beginning, because they must be in the alignments and the SNP data. The outgroup strains must form one clade in the tree; if they do not, EasyCGTree_SpecificSNP.pl stops with a message. Notes:

a. The two branches at the root are one branch of the unrooted tree, so the SNPs that separate the outgroup from the other strains are given on both of them.

b. A very distant outgroup adds many SNP sites and reduces the core sites (the alignments become more gappy); a close outgroup is usually better.

c. If the outgroup strains are also given as 'ignored strains' (Section 7.1), sites that vary only because of them are not SNPs; the counts within the ingroup are then based on the ingroup only, but the root branch gets fewer SNPs.

d. Without an outgroup, the tree is used as it is rooted in the file. The 'strict' mode does not depend on the rooting (only on the branches), but its numbers are still written on one side of each branch.

When another tree is used (`-tree`), it should already be rooted as wanted (e.g. with the outgroup in a tree viewer), or the outgroup is given with `-outgroup` and the tree is rerooted by EasyCGTree_SpecificSNP.pl.

7.3 BuildHMM.pl

Builds a profile HMM database from customized gene clusters (protein sequences, one FASTA file per gene family). The database is written into the PHD (folder 'HMM') and can then be used with '-hmm <name>'.

```
> perl BuildHMM.pl -gc Dir_name [options]
```

-gc <String> (Essential)

Input directory of gene clusters in FASTA format. The file names (without extension) become the gene names.

-name <String>

Name of the HMM database. [default: the name of the input directory]

-force

Overwrite an existing database with the same name.

-aln

The gene clusters are already aligned; BuildHMM.pl will not align them.

-super5 <Int>

Families with at least this number of sequences are aligned with 'muscle -super5'. [default: 1000]

Output: `HMM/<name>.hmm`, `<gc>.BuildHMM_summary.txt` (sequences, alignment length and HMM length of each family) and the working directory `<gc>_BuildHMM_TEM`.

7.4 Gene_Prevelence.pl

Statistics of the prevalence of the genes of a gene set among the genomes, based on the HMM search of EasyCGTree (it is performed with EasyCGTree.pl if it is missing). It also previews how many genomes and genes would pass the cutoffs of EasyCGTree.

```
> perl Gene_Prevelence.pl -input myGenomes [options]
```

-input <String> (Essential)

Input data directory.

-hmm <String> / -evalue <Real>

Profile HMM and E-value [default: those of the existing HMM search].

-force

Run the HMM search again if the existing one was made with another HMM (this replaces the later results in '<input>_TEM').

-genome_cutoff / -gene_cutoff <Decimal>

Cutoffs for the preview of the screening. [default: 0.8]

Output (`<input>.<hmm>.*`): `gene_prevalence.txt` (IDs of the proteins found in each genome), `gene_copies.txt` (copy numbers), `gene_summary.txt` (per gene), `genome_summary.txt` (per genome) and `shared_hits.txt`.

In some genomes, the best hits of two genes can be the same protein (e.g. two domains of a protein that are one CDS in some strains but two CDS in others, or paralogs whose number differs between genomes). EasyCGTree then uses this protein twice, which changes the weight of these genes and adds noise to the phylogenetic signal. This situation is complex, and there is no efficient and rigorous solution yet. In general, it is rare for the genes of widely used gene sets such as bac120 (the relationship of the input genomes also matters), and with many genes its effect on the result is small. These proteins are marked with '*' in the tables, and the pairs of genes are listed in `shared_hits.txt` for reference.

8. Hardware requirement

A normal PC is good enough to run EasyCGTree with limited input data, because HMMER, MUSCLE and FastTree are fast and use several threads on all systems. The speed depends on the size of the input data: with fewer than 100 genomes, a PC finishes the analysis within a few hours based on bac120. For bigger data sets, a powerful PC or server is recommended. Please keep in mind that IQ-TREE takes much more time and memory than FastTree; if a run with IQ-TREE (-tree_app iqtree) fails on a PC, try FastTree, or use a more powerful computer.

9. FAQ

1. EasyCGTree_GUI reports that Perl was not found.

A: Windows users need to install Perl (Section 2.3) and restart EasyCGTree_GUI, or set the path of perl.exe in Settings → Paths and programs.

2. 'The profile HMM bac120 was not found'.

A: The HMM sets are downloaded separately (Section 3.2). Download 'bac120.hmm' (or the set you chose) from the folder 'HMM' of the repository on GitHub or Gitee into the folder 'HMM' of EasyCGTree.

3. A program in 'bin' cannot be started (Linux/macOS).

A: Run `chmod +x bin/*` in the home directory of EasyCGTree. On macOS, also run `xattr -dr com.apple.quarantine .` there once, because macOS blocks programs downloaded from the internet.

4. A later task stops because of the existing results.

A: Each task records its settings in '<input>_TEM/EasyCGTree_record.txt', and later tasks must use the same HMM set, cutoffs and '-outdir'. Run the earlier task again with the new settings, or use '-task all'.

5. A profile HMM built with another HMMER version cannot be read.

A: HMMs built by HMMER 3.1 or newer cannot be read by HMMER 3.0. EasyCGTree 5.0 uses HMMER 3.4, which reads all of them; if older programs were put into 'bin', 'hmmbuild' and 'hmmsearch' must be of the same version.

6. What do the numbers at the branches of the tree mean?

A: They are support values (Section 1.1): FastTree gives SH-like local support (0-1), IQ-TREE with '-B 1000' ultrafast bootstrap (0-100), the consensus tree the percentage of gene trees with the branch, and ASTRAL its own branch support. Branches with low support should be interpreted with care.

7. How is the tree rooted?

A: The trees of EasyCGTree are unrooted; the root shown by a tree viewer is arbitrary. Root the tree with an outgroup in the viewer (e.g. 'Reroot' in FigTree or iTOL), and include suitable outgroup strains in the input data (Section 7.2).

8. Can I use EasyCGTree on a server without a graphical interface?

A: Yes. All functions are available in the Perl scripts (Sections 5 and 7).

10. Citation

Zhang Dao-Feng, He Wei, Shao Zongze, Ahmed Iftikhar, Zhang Yuqin, Li Wen-Jun & Zhao Zhe. EasyCGTree: a pipeline for prokaryotic phylogenomic analysis based on core gene sets. *BMC Bioinformatics* 2023 24:390. https://doi.org/10.1186/s12859-023-05527-2

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